

MAE 3270

Fall 2025

Final HW / Course Project
Part II: Design for Portfolio

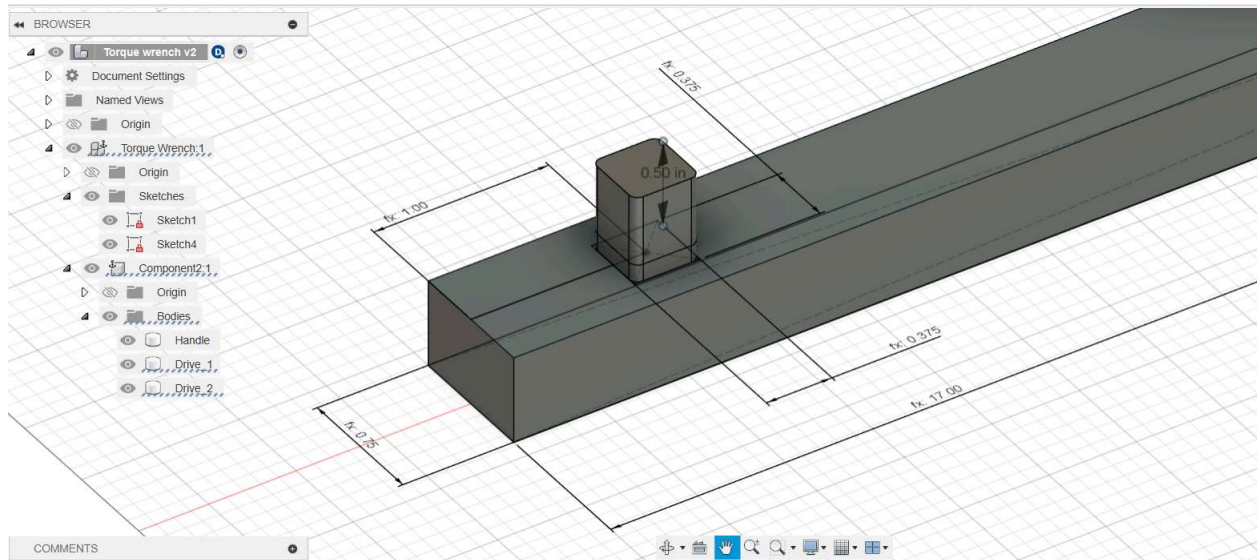
By:

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Summary:

Design a torque wrench of given dimensions that can satisfy factors of safety requirements for brittle fracture, crack propagation, and cyclic loading.

CAD model of torque wrench from Fusion 360:



Material selected for torque wrench:

Titanium, beta alloy, Ti-12Mo-6Zr-2Fe

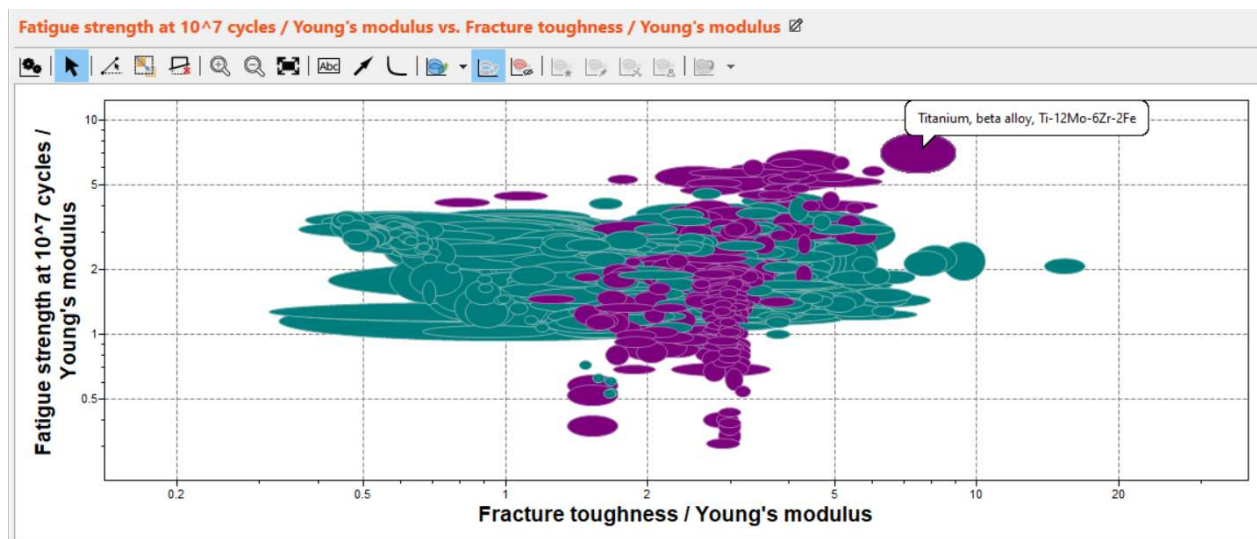
$E = 9.16 - 13.1 \text{ E}6 \text{ psi}$

$\nu = 0.31 - 0.35$

$\sigma_y = 130 - 157 \text{ E}3 \text{ psi}$

Fatigue strength at 10^7 cycles = $69.2 - 86.8 \text{ E}3 \text{ psi}$

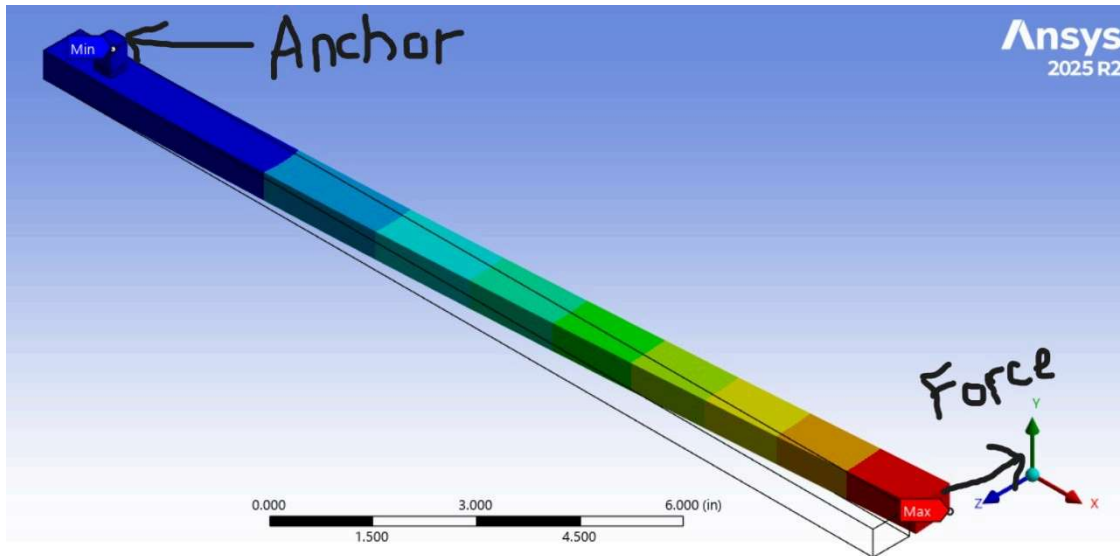
$K_{IC} = 80.1 - 83.7 \text{ E}3 \text{ psi} \cdot \text{in}^{1/2}$



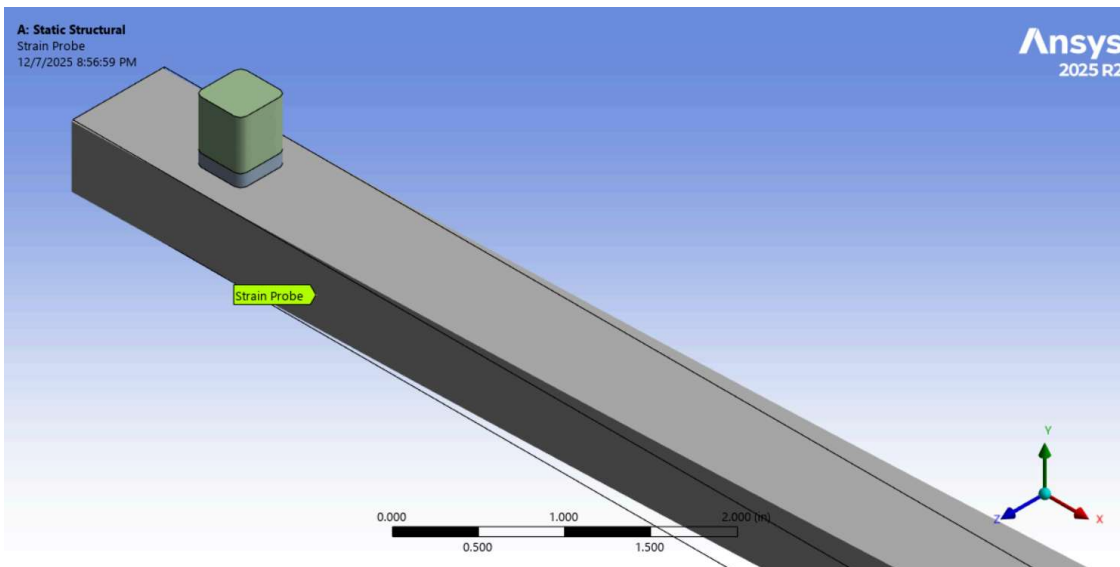
Above shows how this alloy of Titanium compares to other alloys of Titanium as well as alloys of Steel and Aluminum.

Loads and Boundary Conditions, and strain gauge location:

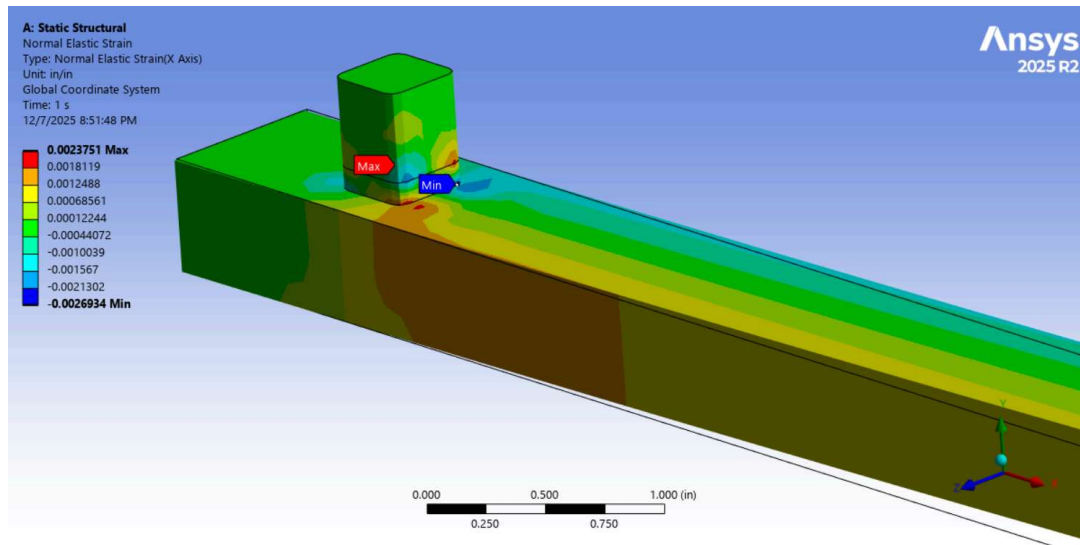
As indicated in the picture below, the tip of the body of the drive was chosen to be the stationary anchor in Ansys. The face of the tip of the handle was chosen to be where the force is applied and thus the maximum deflection.



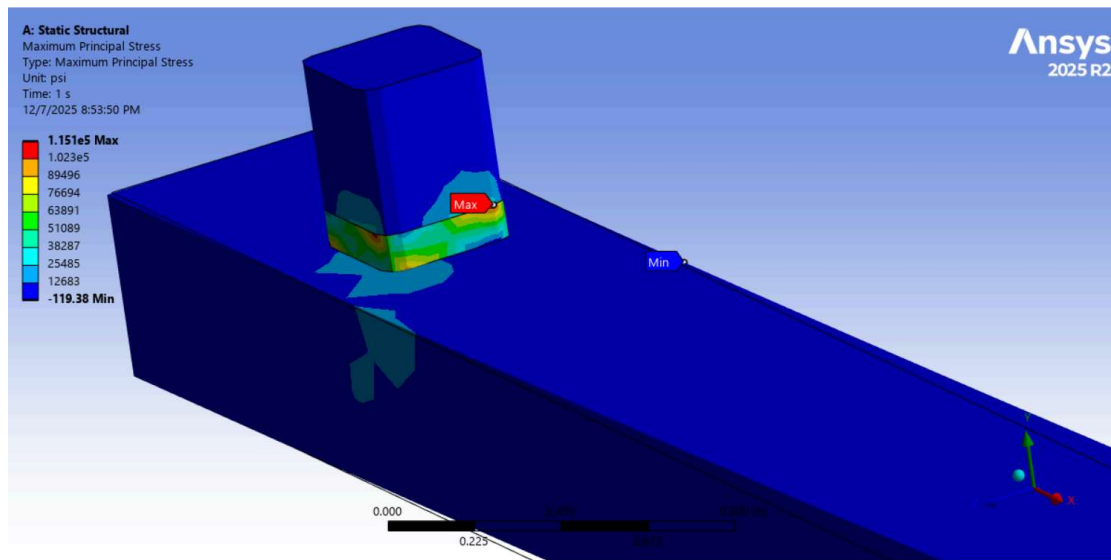
Below is an image showing the location of the strain gauge:



Normal Strain Contours:



Maximum Principal Stress Contours:



Summary of numerical results of FEM:

$\sigma_{\text{Max, Normal}}$: 52,714 psi

ϵ_{Gage} : 1308.4 μ strains

Deflection_{Max}: 0.48383 in

Torque Wrench Sensitivity in mV/V using strain from FEM analysis: 1.3084 mV/V

Strain Gauge Selected: SGT-2LH/350-TY11

We chose a half bridge strain gauge design with a width of 0.158 in and a length of 0.563 in.